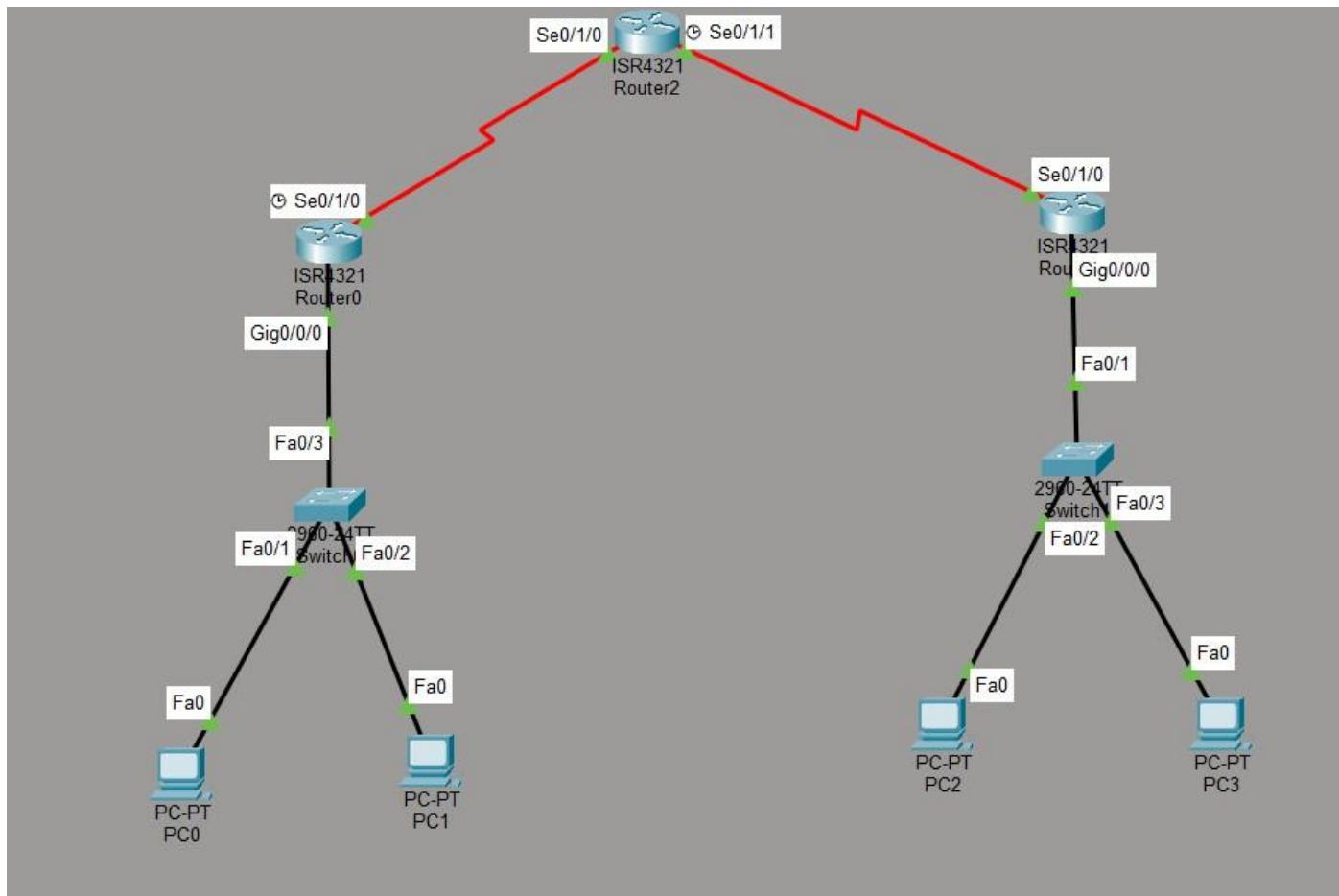


TASKS'

1. Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.



Router 0

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g 0/0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#int s 0/1/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#ip nat outside
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router(config)#ip address 10.0.0.4 255.255.255.252 10.0.0.2
^
% Invalid input detected at '^' marker.

Router(config)#ip route 10.0.0.4 255.255.255.252 10.0.0.2
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

```

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config)#access-list 100 permit ip 192.168.10.0 0.0.0.255 any
Router(config)#ip nat inside source list 100 interface serial 0/1/0 overload
Router(config)#do wr
Building configuration...
[OK]
Router(config)#access-list 110 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.6
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.6
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 110
Router(config-crypto-map)#exit
Router(config)#interface serial 0/1/0
Router(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

```

Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA

```

dst	src	state	conn-id	slot	status
10.0.0.6	10.0.0.1	QM_IDLE	1058	0	ACTIVE

Router 1

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g 0/0/0/
^
% Invalid input detected at '^' marker.

Router(config)#int g 0/0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#int s 0/1/0
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#ip nat outside
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#
Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router(config)#ip route 10.0.0.0 255.255.255.252 10.0.0.5
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.5
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.5
Router(config)#do wr
Building configuration...
[OK]

Router(config)#access-list 100 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config)#access-list 100 permit ip 192.168.20.0 0.0.0.255 any
Router(config)#ip nat inside source list 100 interface serial 0/1/0 overload
Router(config)#do wr
Building configuration...
[OK]

Router(config)#access-list 110 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 110
Router(config-crypto-map)#exit
Router(config)#interface serial 0/1/0
Router(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

Router 2

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int s 0/1/1
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

Router(config)#int s 0/1/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router(config)#int loopback 0

Router(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

Router(config-if)#ip address 8.8.8.8 255.255.255.255
Router(config-if)#exit
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.6
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

PC 0 Ping

```
C:\>ping 192.168.20.10

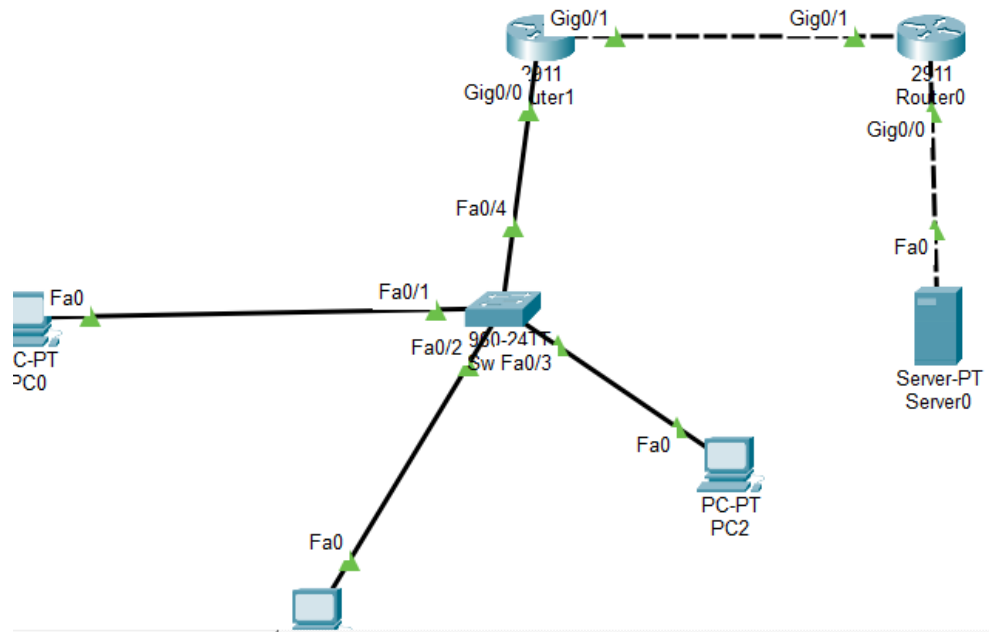
Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=19ms TTL=126
Reply from 192.168.20.10: bytes=32 time=12ms TTL=126
Reply from 192.168.20.10: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 19ms, Average = 11ms

C:\>
```

2. Configure Port Address Translation (PAT) on a router in your simulated WAN so that three devices from the internal network can browse the internet using a single public IP address. **Hint:** Use the 'overload' keyword in your NAT configuration.



Router 1

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 200.0.0.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#ip nat inside
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 200.0.0.2 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#access-list 1 permit 10.0.0.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface gigabitEthernet 0/1 overload
Router(config)#ip route 0.0.0.0 0.0.0.0 200.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
```

Pc1

```
C:\>ping 30.0.0.10

Pinging 30.0.0.10 with 32 bytes of data:

Request timed out.
Reply from 30.0.0.10: bytes=32 time=1ms TTL=126
Reply from 30.0.0.10: bytes=32 time<1ms TTL=126
Reply from 30.0.0.10: bytes=32 time<1ms TTL=126

Ping statistics for 30.0.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

R1# show ip nat translations

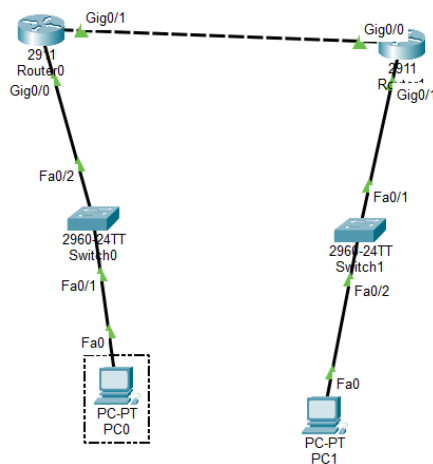
R1# show ip nat statistics

```

Router#show ip nat statistics
Total translations: 3 (1 static, 1 dynamic, 1 extended)
Outside Interfaces:
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/1
Hits: 0 Misses: 0
Expired translations: 3
Dynamic mappings:
Router#

```

3. Intentionally misconfigure the VPN tunnel's pre-shared key between two routers in your WAN simulation, then use the 'show crypto isakmp sa' and 'show run' commands to identify and fix the connectivity issue.



Router 1

```

Router>ena
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#exit
Router(config)#
Router(config)#crypto isakmp key cisco123 address 20.0.0.2
Router(config)#access-list 100 permit ip 10.0.0.0 0.0.0.255 30.0.0.0 0.0.0.255
Router(config)#
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 20.0.0.2
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#
Router(config)#interface g0/1
Router(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

```

```

Router>ena
Router#show crypto isakmp sashow crypto isakmp sa^Z
Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
20.0.0.2      20.0.0.1      MM_NO_STATE      0      0 ACTIVE (deleted)

```

IPv6 Crypto ISAKMP SA

Router 2

```

Router>
Router>
Router>
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#exit
Router(config)#crypto isakmp key wrong123 address 20.0.0.1
Router(config)#
Router(config)#access-list 100 permit ip 30.0.0.0 0.0.0.255 10.0.0.0 0.0.0.255
Router(config)#
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router(config-crypto-map)#set peer 20.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#
Router(config)#interface g0/1
Router(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#

```

Wrong key fix karo

R2:

```

Router#
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no crypto isakmp key wrong123 address 20.0.0.1
Router(config)#crypto isakmp key cisco123 address 20.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

4. Troubleshoot a scenario where devices in one branch office can ping the internet but cannot access devices in the other branch via the VPN tunnel; document the commands you use and the configuration change that resolves the problem.



Scenario and Topology

Branch A

Branch B

PC0/PC1/PC2 — Switch — Router1 ===== WAN ===== Router0 — Switch — PC3

192.168.1.0/24 200.1.1.2 200.1.1.1 192.168.2.0/24

Router1 represents Branch A and Router0 represents Branch B. The serial WAN connection between the routers is operational. Internet connectivity from Branch A is working, but traffic destined for the Branch B LAN initially fails through the VPN.

IP Addressing

Device	Interface	IP Address	Role
Router1	G0/0/0	192.168.1.1/24	Branch A gateway
Router1	S0/1/0	200.1.1.2/30	WAN / VPN peer
Router0	S0/1/0	200.1.1.1/30	WAN / VPN peer
Router0	G0/0/0	192.168.2.1/24	Branch B gateway
PC0	Fa0	192.168.1.10/24	Branch A host
PC1	Fa0	192.168.1.11/24	Branch A host
PC2	Fa0	192.168.1.12/24	Branch A host
PC3	Fa0	192.168.2.10/24	Branch B host

Problem Statement

The branch network could reach the Internet, confirming that the basic LAN, WAN, routing, and Internet path were functioning. However, a host in Branch A could not reliably access the Branch B host at 192.168.2.10 through the intended IPsec VPN.

PC0> ping 8.8.8.8

PC0> ping 192.168.2.10

The Internet ping was successful while the branch-to-branch test was unsuccessful. This indicated that the fault was specific to the VPN path or the traffic selected for encryption.

Troubleshooting Commands

Check ISAKMP Status

Router1# show crypto isakmp sa

This command was used to check whether the ISAKMP security association was established. An established tunnel is commonly shown with a state such as QM_IDLE.

Check IPsec Security Associations

```
Router1# show crypto ipsec sa
```

The IPsec SA output was checked for packet counters, especially #pkts encaps and #pkts decaps. These counters help determine whether traffic is actually entering and leaving the IPsec tunnel.

Check Access Lists

```
Router1# show access-lists
```

The VPN crypto ACL was inspected because an IPsec crypto map uses an ACL to identify the interesting traffic that must be encrypted.

Check the Running Configuration

```
Router1# show run | section access-list
```

The running configuration was reviewed to compare the configured VPN traffic selectors with the actual Branch A and Branch B networks.

Fault Identified

The problem was an incorrect destination network in Router1's crypto ACL. The actual Branch B network was 192.168.2.0/24, but the ACL incorrectly referenced 192.168.3.0/24.

Incorrect configuration:

```
access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

Because 192.168.2.0/24 did not match the crypto ACL, the branch-to-branch traffic was not selected as IPsec interesting traffic.

Configuration Change / Solution

The incorrect ACL was removed and replaced with the correct Branch A-to-Branch B selector:

```
Router1# configure terminal
```

```
Router1(config)# no access-list 110
```

```
Router1(config)# access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
```

```
Router1(config)# end
```

The corrected ACL now matches:

192.168.1.0/24 → 192.168.2.0/24

Verification

After correcting the ACL, the branch-to-branch connectivity test was repeated:

```
PC0> ping 192.168.2.10
```

Successful replies confirmed that Branch A could communicate with the Branch B host.

The VPN was then verified with:

Router1# show crypto isakmp sa Router1# show crypto ipsec sa

The ISAKMP state should be established (commonly QM_IDLE), and the IPsec encapsulation and decapsulation counters should increase when branch-to-branch traffic is generated.

Result

The connectivity problem was successfully diagnosed and resolved. Internet access remained available, while the branch-to-branch VPN traffic initially failed because Router1's crypto ACL specified the wrong destination network. Updating the ACL to permit traffic from 192.168.1.0/24 to 192.168.2.0/24 restored communication between the two branch offices.

- 5. Take the WAN, VPN, and NAT scenario you built and swap your configuration file with a classmate; review their setup, find one misconfiguration, and write a short note on how you identified and fixed it.

Constraint: Focus on using 'show' and 'debug' commands for troubleshooting rather than guessing the issue.**



WAN, VPN and NAT Troubleshooting

Practical Objective

Review a classmate's WAN, VPN, and NAT configuration, identify one misconfiguration using show and debug commands, and fix the issue.

Troubleshooting Procedure

1. Check Interface Status

Used to verify IP addresses and confirm that the WAN and LAN interfaces are up/up.

```
Router# show ip interface brief
```

2. Check Routing Table

Used to verify that the required networks and routes are present.

```
Router# show ip route
```

3. Check NAT

Used to verify whether NAT translations are being created and to inspect NAT operation.

```
Router# show ip nat translations
```

```
Router# show ip nat statistics
```

4. Check VPN Status

Used to verify the ISAKMP/IPsec security associations and VPN traffic.

```
Router# show crypto isakmp sa
```

```
Router# show crypto ipsec sa
```

5. Use Debug Commands

Debug output was observed while generating test traffic to identify where the connection was failing.

```
Router# debug ip nat
```

```
Router# debug crypto isakmp
```

```
Router# debug crypto ipsec
```

6. Stop Debugging

Used to safely stop all active debugging commands.

```
Router# undebug all
```

Misconfiguration Found

The LAN interface was not configured as an inside NAT interface. Because of this, NAT translations were not being created for LAN traffic.

Fix Applied

```
Router(config)# interface g0/0
```

```
Router(config-if)# ip nat inside
```

The WAN interface was verified as the NAT outside interface:

```
Router(config)# interface g0/1
```

```
Router(config-if)# ip nat outside
```

Verification

After applying the fix, a ping was generated from the LAN PC to the remote network. The NAT operation was then verified using:

```
Router# show ip nat translations Router# show ip nat statistics
```